Probabilistic Systems Analysis and Applied Probability
Personal Notes

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Contents

1 Probability Models and Axioms

Probability is a mathematical framework for reasoning about uncertainty. Its value is that it replaces vague talk about what is likely with a structure rigid enough to compute in, and the structure is remarkably small: every probabilistic model consists of exactly two ingredients. The first says what might happen. The second says how likely each possibility is. This lecture builds both, states the axioms any assignment of likelihood must obey, and works the two settings — discrete and continuous — that the rest of the course returns to constantly.

1.1 The Sample Space

Definition 1.1 (Sample space). The *sample space* Ω of an experiment is the set of its possible outcomes.

The list of outcomes is not arbitrary. It must be mutually exclusive, so that exactly one outcome occurs, and collectively exhaustive, so that at least one of them does. Together these say that when the experiment is run, exactly one element of Ω is realized.

Correctness leaves a great deal of freedom, and choosing well is a matter of taste rather than rule. The sample space must be at the right granularity: fine enough to express every event of interest, coarse enough to leave out detail that plays no part in the analysis. A model of a coin toss that records the coin's temperature is not wrong, only useless.

Example: Roll a tetrahedral (four-sided) die twice. Taking the outcome to be the ordered pair of face values,

$$\Omega = \{(x, y) : x, y \in \{1, 2, 3, 4\}\},$$

which has 16 elements. The same experiment admits a sequential description as a tree: a first branching into the four values of x, and below each of those a second branching into the four values of y. The tree and the 4×4 grid describe the same sample space; the tree merely emphasizes that the rolls happen in order.

A sample space need not be finite, or even countable.

Example: Pick a point from the unit square. Here

$$\Omega = \{(x, y) : 0 \leq x \leq 1,\ 0 \leq y \leq 1\},$$

an uncountably infinite set of outcomes.

1.2 Events and the Axioms

Definition 1.2 (Event). An *event* is a subset of the sample space.

Probability is assigned to events rather than to outcomes. This is not pedantry. In the unit square every individual outcome will turn out to have probability zero, so a model that spoke only of outcomes would say nothing at all; it is the events, sets like "the point lies in the left half," that carry the content.

Definition 1.3 (Probability law). A *probability law* assigns to each event A a number $\mathbb{P}(A)$, the probability of A, subject to three axioms:

1. *Nonnegativity*: $\mathbb{P}(A) \geq 0$ for every event A.
2. *Normalization*: $\mathbb{P}(\Omega) = 1$.
3. *Additivity*: if $A \cap B = \emptyset$, then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.

Everything else is derived. Applying additivity repeatedly to a finite collection of distinct outcomes $s_1, \ldots, s_k$ gives

$$\mathbb{P}(\{s_1, s_2, \ldots, s_k\}) = \mathbb{P}(\{s_1\}) + \cdots + \mathbb{P}(\{s_k\}) = \mathbb{P}(s_1) + \cdots + \mathbb{P}(s_k),$$

writing $\mathbb{P}(s)$ for $\mathbb{P}(\{s\})$ as is customary. So on a finite sample space a probability law is completely determined by the probabilities of the individual outcomes.

Note: *Two loose ends, both settled below or later. First, Axiom 3 covers two disjoint events and hence, by induction, any finite number — but not infinitely many, and that is not enough. Second, on a continuous sample space one cannot assign a probability to* every *subset without contradiction; sufficiently pathological sets admit no consistent value. The axioms are therefore imposed on a restricted family of events, which includes everything built from intervals and rectangles by countably many set operations. Every event in this course is of that kind.*

1.3 A Probability Law on a Finite Sample Space

Return to the two rolls of a tetrahedral die, and let each of the 16 outcomes have probability 1/16. Since the outcome probabilities determine the law, every event can now be computed by counting cells.

- $\mathbb{P}\big((X, Y) \text{ is } (1,1) \text{ or } (1,2)\big) = \frac{2}{16} = \frac{1}{8}$, two cells.
- $\mathbb{P}(X = 1) = \frac{4}{16} = \frac{1}{4}$, the column of four cells above $X = 1$.
- $\mathbb{P}(X + Y \text{ is odd}) = \frac{8}{16} = \frac{1}{2}$: the sum is odd exactly when one roll is odd and the other even, which happens in $2 \cdot 2 + 2 \cdot 2 = 8$ cells.
- $\mathbb{P}(\min(X, Y) = 2) = \frac{5}{16}$: the minimum is at least 2 in the $3 \times 3 = 9$ cells with both rolls in $\{2, 3, 4\}$ and at least 3 in the $2 \times 2 = 4$ cells with both in $\{3, 4\}$, leaving $9 - 4 = 5$.

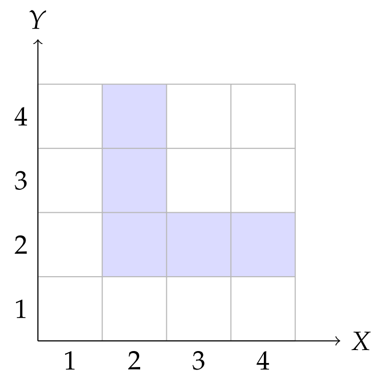

Figure 1: The sample space for two rolls of a tetrahedral die, with X the first roll and Y the second. The five shaded cells are the event $\{\min(X, Y) = 2\}$.

1.4 The Discrete Uniform Law

The die example used no property of the number 16 beyond the fact that the outcomes were equally likely, and that special case deserves a name.

Theorem 1.4 (Discrete uniform law)**.** *If Ω is finite and all outcomes are equally likely, then for every event A,*

$$\mathbb{P}(A) = \frac{\textit{number of elements of } A}{\textit{total number of sample points}}.$$

Note: *Under the uniform law, computing probabilities* is *counting. This is why a later lecture is given over entirely to counting techniques, and it is what gives precise meaning to the phrases "fair coin," "fair die," and "well-shuffled deck": each names a uniform law on the appropriate sample space.*

1.5 The Continuous Uniform Law

Now take two "random" numbers in $[0, 1]$, meaning a point drawn from the unit square with no region favored over another of the same size. The uniform law here assigns each event its area:

$$\mathbb{P}(A) = \text{area of } A.$$

This satisfies the axioms: areas are nonnegative, the unit square has area 1, and disjoint regions have areas that add.

Example: $\mathbb{P}(X + Y \leq 1/2)$ is the area of the triangle with vertices $(0, 0)$, $(1/2, 0)$, and $(0, 1/2)$, namely $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$.

Example: $\mathbb{P}\big((X, Y) = (0.5,\ 0.3)\big) = 0$, since a single point has zero area.

The second example is worth pausing on. Every individual outcome has probability zero, yet one of them occurs. Probability zero does not mean impossible, and on a continuous sample space the outcomes carry no information whatever — only the events do.

1.6 Countable Additivity

The gap left by Axiom 3 appears as soon as the sample space is countably infinite.

Example: Let $\Omega = \{1, 2, \ldots\}$ with $\mathbb{P}(n) = 2^{-n}$ for each n. What is the probability that the outcome is even?

The natural computation sums a geometric series:

$$\mathbb{P}(\{2, 4, 6, \ldots\}) = \mathbb{P}(2) + \mathbb{P}(4) + \cdots = \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \cdots = \frac{1/4}{1 - 1/4} = \frac{1}{3}.$$

But this breaks an infinite union into infinitely many pieces, and finite additivity licenses no such step. The axiom must be strengthened.

Definition 1.5 (Countable additivity). If $A_1, A_2, \ldots$ is a sequence of pairwise disjoint events, then

$$\mathbb{P}(A_1 \cup A_2 \cup \cdots) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \cdots .$$

Countable additivity replaces Axiom 3 and implies it, taking all but two of the events empty. Countable is also as far as one can go: additivity over an *uncountable* family is false, as the unit square already shows. There the whole square is the disjoint union of its individual points, each of probability zero, and no sum of zeros gives 1.

Source: MIT 6.041 Probabilistic Systems Analysis and Applied Probability, Fall 2010, Lecture 1 notes by John Tsitsiklis.